Inflatable Ceramic Water Heater (CBPC)

Cost Analysis per Unit — CSMFAB 006

Category: Aegis Home — Plumbing **Unit Basis:** 1 standard 40-gallon unit **Date:** June 2026 | **Document Type:** CSMFAB Cost Study V2.0

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1. Product Summary

This cost analysis is derived from the V2.0 specification and v1to2 versioning document for this product. All material costs reflect 2026 market pricing. Three production scales are costed: - **Prototype:** 1-5 units (single-setup, high labor, no tooling amortization) - **Small Batch:** 100-500 units/year (semi-tooled, moderate labor) - **Production Volume:** 5,000-50,000 units/year (fully tooled, optimized labor)

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2. Bill of Materials (BOM)

Component	Quantity/Unit	Unit Price	Extended Cost
CBPC (chem-bonded phosphate ceramic) powder mix (50 kg bag)	50 kg	\$4.50/kg	\$225.00
Inflatable HDPE inner formwork bladder	1.5 kg	\$8.00/kg	\$12.00
Si_3N_4 resistive heating element (3 kW)	0.8 kg	\$28.00/kg	\$22.40
Zirconia-phosphate foam shell (outer, 50mm)	8 kg	\$15.00/kg	\$120.00
PEX-a inlet/outlet fittings (ZrO_2 ceramic)	0.4 kg	\$45.00/kg	\$18.00
Aerogel thermal insulation wrap (25mm)	0.5 m^2	\$250.00/kg	\$125.00
PEEK CF40 pressure relief valve	0.15 kg	\$280.00/kg	\$42.00
$KNbO_3$-$BaTiO_3$ temperature sensor	0.01 kg	\$180.00/kg	\$1.80
BOM Subtotal			**\$566.20**

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3. Labor, Process & Overhead Costs

Cost Element	Prototype	Small Batch	Production Volume
BOM Materials	\$566.20	\$464.28	\$368.03
Direct Labor	\$280.00	\$180.00	\$120.00
Overhead (15% of labor)	\$42.00	\$27.00	\$18.00
Quality Control & Testing	\$33.60	\$14.40	\$6.00
Packaging & Logistics	\$25.00	\$20.00	\$15.00

Tooling & Equipment Notes: One-time inflatable mold: $400. CBPC cast process: low equipment cost.

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4. Per-Unit Cost Summary

Scale	Total COGS	Suggested MSRP Range	Gross Margin
Prototype (1-5 units)	**$1,020**	$1,632 – $2,244	35-55%
Small Batch (100-500/yr)	**$680**	$1,020 – $1,360	33-50%
Production Volume (5K-50K/yr)	**$480**	$672 – $864	28-45%

Market Reference: $600-900 (vs. $400-800 standard water heater; GIC immunity is differentiation)

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5. Key Cost Drivers (V2.0)

Driver	Impact	Mitigation
ZrB2/Si3N4 UHTC ceramics	HIGH — specialty powder $350+/kg	Bulk procurement; domestic synthesis pilot (Japan NIMS CO2 route)
MXene Ti3C2Tx	VERY HIGH — $2,000/kg at 2026 scale	Tile pattern minimizes usage; price projected -60% by 2028 as production scales
Elium® thermoplastic BFRP	MEDIUM — recyclability adds value; $12/kg resin	Phoenix Protocol recycling recovers 100% monomer; net cost offset
Tooling NRE	HIGH for first production run	Paid by pre-orders or grant funding; amortized quickly
Flash Sintering process	MEDIUM — shares SPS equipment	Regional ceramic sintering facility sharing model

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6. Phoenix Protocol Circular Economy Value

At end-of-life, material recovery adds back-value: - **Recovered material value per unit:** $101 – $198 - **Recovery reduces net lifecycle cost by:** 15-30% - **Critical minerals (In, Y, Co, Ti) recovered** via ionic liquid extraction

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End of Inflatable Ceramic Water Heater (CBPC) Cost Analysis | CSMFAB 006 | June 2026